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Basic Possibilistic Logic: Examples and Visualization

Seminar Paper
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Abstract

Possibilistic logic is a non-classical logic framework designed to handle uncertainty by incorporating degrees of possibility. This paper provides a general introduction to possibilistic logic by using strong examples of the apllication.

1 Introduction

Reasoning under uncertainty is a significant problem in artificial intelligence. Several approaches have been proposed to address this issue, including possibility theory [8], fuzzy logic [13], and possibilistic logic [1]. The foundation of possibility theory was laid by Zadeh in 1978 [14] as an extension of fuzzy set theory [13], aiming to provide a qualitative framework for handling uncertain information. Building upon Zadeh's work, Dubois and Prade formulated possibilistic logic [1] to integrate possibility theory into formal logic systems.
Possibilistic logic has been further developed for diverse applications such as default reasoning, belief revision, or decision-making under uncertainty with generalized approaches; see [6]. The goal of this paper is to introduce basic possibilistic logic with examples and visualizations. The focus is on providing examples with additional explanations and graphical representations. In this paper, we will focus solely on basic possibilistic logic.
This work is structured as follows. In Section 2, basic possibilistic logic is explained, including its preliminaries for understanding the syntax and semantics of the logic. Before the core part of this paper, Section 3 compares possibilistic logic with classical logic, and Section 4 discusses its relationship with artificial intelligence. Section 5 illustrates the application of basic possibilistic logic through an example and graphical explanations. Finally, Section 6 summarizes and discusses the results of the previous sections.

2 Fundamentals of Basic Possibilistic Logic

The following introduction to basic possibilistic logic is inspired by the presentation of Dubois and Prade [2]. This section provides a brief overview of basic possibilistic logic. For more details, we refer to the works [2] and [1].
In this section, we first introduce the main concepts of possibility theory, which are necessary for understanding possibilistic logic. Then, we present basic possibilistic logic, covering its syntax, axioms, inference rules, and semantic aspects.

2.1 Possibility Theory

Possibility theory is explained in detail in [5] and [14]. It relies on the concept of possibility distributions.

Definition 1 *A possibility distribution is a function π that maps elements from a set of states U to a totally ordered scale $[0, 1]$, see [2]:*

$$\pi : U \to [0, 1] .$$

The value $\pi(u)$ indicates the degree of possibility assigned to the state u. This function represents the agent's knowledge and helps differentiate between states that are more or less plausible. A state u is considered impossible if $\pi(u) = 0$, whereas it is fully possible if $\pi(u) = 1$.

Example 1 *Consider a weather forecasting scenario where an agent assesses the possibility of different weather conditions for tomorrow. Let $U = \{sunny, cloudy, rainy, stormy\}$ be the set of possible states. The agent assigns a possibility distribution π as follows:*
$\pi(sunny) = 1, \quad \pi(cloudy) = 0.7, \quad \pi(rainy) = 0.3, \quad \pi(stormy) = 0.$
The agent is completely certain that a sunny day is fully possible ($\pi(sunny) = 1$), considers cloudy weather strong possible ($\pi(cloudy) = 0.7$), finds rain unlikely ($\pi(rainy) = 0.3$), and states a storm impossible ($\pi(stormy) = 0$).

The basic measures of possibility theory model the possibility and necessity of events (see [2], [10]). If the occurrence of an event A is of interest, possibility theory provides two measures derived from the possibility distribution of A.

Definition 2 *The possibility of an event A is defined as:*

$$\Pi(A) = \sup_{u \in A} \pi(u) .$$

The possibility measure represents the maximum possibility assigned to any state in A. It evaluates the extent to which A is consistent with π. Possibility measures satisfy the characteristic maxitivity property:

$$\Pi(A \cup B) = \max(\Pi(A), \Pi(B)) .$$

Definition 3 *The necessity of an event A is defined as:*

$$N(A) = \inf_{u \notin A} (1 - \pi(u)).$$

The necessity measure represents the minimum of the inverted possibility values of all states not in A. It evaluates the extent to which A is certainly implied by π. Necessity measures satisfy the characteristic minimization property:

$$N(A \cap B) = \min(N(A), N(B)).$$

This property expresses that for $A \cap B$ to be certain, both A and B must be certain individually. The duality between these two measures is given by:

$$N(A) = 1 - \Pi(A^c),$$

where A^c is the complement of A.

Example 2 *Consider a possibility distribution π over a set of possible weather conditions: $U = \{sunny, cloudy, rainy\}$ with the values $\pi(sunny) = 1$, $\pi(cloudy) = 0.7$, and $\pi(rainy) = 0.3$.*
The possibility of the event "not rainy" $A = \{sunny, cloudy\}$ is given by:

$$\Pi(A) = \sup_{u \in A} \pi(u) = \max(\pi(sunny), \pi(cloudy)) = \max(1, 0.7) = 1.$$

The necessity of "not rainy" is computed as:

$$N(A) = \inf_{u \notin A}(1 - \pi(u)) = 1 - \pi(rainy) = 1 - 0.3 = 0.7.$$

This means that the event "not rainy" is fully possible but not completely necessary.

2.2 Formal Foundations of Basic Possibilistic Logic

This section lays out the foundations of basic possibilistic logic. First, the syntax is introduced. Second, the semantic aspects of the logic are derived. Third, the inference rules are explained. Finally, the axioms of this logic are covered. Further references for this section are [2], [9].

Definition 4 *The syntax of a basic possibilistic logic formula is:*

$$(a, \alpha)$$

where a is a classical logic formula (e.g., $p \to q$) and α is the certainty level (the minimum necessity degree of a). The classical logic formula can be from propositional logic or first-order logic (see [2]).

The semantics of a possibilistic logic formula are based on the semantics of the corresponding classical logic formula. The pair (a, α) is interpreted as $N(a) > \alpha$, meaning that the formula a is at least certain to degree α. The higher the value of α, the stronger the belief in a.

Example 3 $(b \to f, 0.9)$ *means that "birds fly" holds with certainty* 0.9.

The following inference rules of basic possibilistic logic are necessary for the application of the logic in this paper. The *certainty weakening rule* states that if a formula holds with a certainty level α, it also holds with any lower certainty level β because we are being less demanding, see Equation 1.

$$\text{If } \beta \leq \alpha, \quad (a, \alpha) \vdash (a, \beta) \tag{1}$$

The *modus ponens rule* states that if $a \to b$ holds with certainty level α, and a also holds with certainty level α, then b must hold with certainty level α, see Equation 2.

$$(\neg a \vee b, \alpha), (a, \alpha) \vdash (b, \alpha), \quad \forall \alpha \in (0, 1] \tag{2}$$

In classical logic, the *resolution rule* allows us to combine two clauses that share a literal, eliminating that literal and inferring a new clause. In possibilistic logic, the resolution rule preserves the certainty level α, since the inferred formula $b \vee c$ cannot be more certain than the least certain premise, see Equation 3.

$$(\neg a \vee b, \alpha), \quad (a \vee c, \alpha) \vdash (b \vee c, \alpha) \tag{3}$$

The *weakest link resolution rule* extends the resolution rule by considering different certainty levels α and β. It reflects the idea that the certainty of the conclusion cannot exceed the weakest certainty of the premises, see Equation 4.

$$(\neg a \vee b, \alpha), \quad (a \vee c, \beta) \vdash (b \vee c, \min(\alpha, \beta)) \tag{4}$$

For inference with basic possibilistic logic, a possibilistic logic base is used.

Definition 5 *A possibilistic knowledge base is a collection of possibilistic formulas:*

$$\Gamma = \{(a_i, \alpha_i), i = 1, \ldots, m\} \tag{5}$$

Example 4 *Consider a medical diagnosis system where symptoms are associated with diseases with varying degrees of certainty. The possibilistic knowledge base is given by:*

$$\Gamma = \{(\mathit{Fever} \Rightarrow \mathit{Flu}, 0.9), (\mathit{Cough} \Rightarrow \mathit{Cold}, 0.7), (\mathit{Headache} \Rightarrow \mathit{Migraine}, 0.8)\}.$$

This means that having a fever strongly suggests flu with a certainty level of 0.9, a cough suggests a cold with 0.7, and a headache suggests a migraine with 0.8.

The α-cut of a knowledge base contains only those formulas with a certainty level of at least α. It has a nested structure where Γ_α is a subset of Γ_β if $\alpha \geq \beta$.

Definition 6 *The α-cut of a knowledge base is defined as:*

$$\Gamma_\alpha = \{(a_i, \alpha_i) \in \Gamma \mid \alpha_i \geq \alpha\} \tag{6}$$

Example 5 *Continuing with Example 4, the α-cut of the knowledge base for $\alpha = 0.8$ is:*

$$\Gamma_\alpha = \{(\mathit{Fever} \Rightarrow \mathit{Flu}, 0.9), (\mathit{Headache} \Rightarrow \mathit{Migraine}, 0.8)\}.$$

A possibilistic inference (a, α) depends only on formulas with certainty levels $\geq \alpha$, as shown in Equation 7:

$$\Gamma \vdash (a, \alpha) \iff \Gamma_\alpha \vdash (a, \alpha) \tag{7}$$

An important property of knowledge bases is their inconsistency level.

Definition 7 *The inconsistency level of a knowledge base Γ is defined as:*

$$\mathit{inc}\text{-}l(\Gamma) = \max\{\alpha \mid \Gamma \vdash (\bot, \alpha)\} \tag{8}$$

where $\bot$ represents an inconsistent state in logic.

It represents the highest certainty level at which the knowledge base becomes inconsistent. Any formula with a certainty level higher than the inconsistency level remains free from inconsistency.

Example 6 *In the above example are no conflicting statements (e.g., one rule stating "Fever implies Flu" and another stating "Fever does not imply Flu"), there is no contradiction in this knowledge base. The inconsistency level inc-$l(\Gamma)$ is 0, meaning the knowledge base is fully consistent at all certainty levels.*

3 Differences from classical logic

Possibilistic logic extends classical logic with incorporating a certainty degree to classical logic formulas, see [2]. In classical logic a statement is either true or false corresponding to a binary value system. In contrast in possibilistic logic the certainty degree of the possbilistic formula is a non-binary value [1].
In classical logic the presence of a contradiction $(a \wedge \neg a)$ renders the entire knowledge base invalid, allowing any formula to be derived. Possibilistic logic can handle inconsistency. The inconsistency degree of a knowledge base reflects the extent to which it is contradictory [2]. This feature makes it robust for reasoning in incomplete and inconsistent environments.

4 Relation to Artificial Intelligence

Possibilistic logic has with its strength in modeling certainty and handling inconsistency different aplication fields of artificial intelligence problems [2], [4]. In default reasoning possibilistic logic handles incomplete information reasoning. Possibilistic logic can be applied in information fusion where knowledge is combined from different sources. The logic can help with decisions under uncertainty. One application is the usage in medical diagnosis problems [7]. In [3], [11] connections between reasoning with possibilistic logic and machine learning are shown.

5 Analysis and Development of Examples

In the following we will go through an example showing the application of basic possibilistic logic. One of the classical examples in non-monotonic reasoning is the penguin problem. Basic possibilistic logic provides a structured way to handle exceptions of general rules like "birds fly" by associating rules with certainty levels and defining a preference ordering over possible worlds. Each possible world is a combination of the language interpretation of the underlying rules, see [2]. This is example can be found in shorter notice in [2], [12].

5.0.1 Default Rules

The knowledge base of this example consists of three default rules about birds and penguins. A default rule expresses general knowledge and holds in most cases, but it may have exceptions:

$$
\begin{aligned}
&d_1 : b \to f \quad (\text{"Birds fly"}),\\
&d_2 : p \to \neg f \quad (\text{"Penguins do not fly"}),\\
&d_3 : p \to b \quad (\text{"Penguins are birds"}).
\end{aligned}
$$

In the following the goal is to assign certainty levels to these rules, to get a possibilistic knowledge base. In order to find this knowledge base, the following steps are necessary. First the constraints from the rules are constructed. Second, the possible worlds are ranked by the constraints. With the ranked worlds the certainty of the rules is calculated.

5.0.2 Constraints

From the default rules, a set of comparative possibility constraints can be constructed. Comparative possibility constraints are derived from the expression that a rule $a \to b$ gives situations where a and b is true a higher possibilty than that b is false, see [2]:

$$
\begin{aligned}
&C_1 : b \wedge f >_\Pi b \wedge \neg f \quad (\text{birds fly}),\\
&C_2 : p \wedge \neg f >_\Pi p \wedge f \quad (\text{penguins do not fly}),\\
&C_3 : p \wedge b >_\Pi p \wedge \neg b \quad (\text{penguins are birds}).
\end{aligned}
$$

For example, C_1 states that the worlds where birds fly are more possible than worlds where birds do not fly. These constraints establish an ordering over possible worlds, which will be illuminated next.

5.0.3 Possible Worlds

The propositional language for this example involves three propositions: b (bird), p (penguin), and f (flies). The set of possible worlds, Ω, consists of all possible combinations of (b, f, p) represented by ω_i:

$$\Omega = \{\omega_0, \omega_1, \omega_2, \omega_3, \omega_4, \omega_5, \omega_6, \omega_7\}.$$

World	b (Bird)	f (Flies)	p (Penguin)	d_1: $b \to f$	d_2: $p \to \neg f$	d_3: $p \to b$
ω_0	0	0	0			
ω_1	0	0	1		satisfied	violated
ω_2	0	1	0			
ω_3	0	1	1		violated	violated
ω_4	1	0	0	violated		
ω_5	1	0	1	violated	satisfied	satisfied
ω_6	1	1	0	satisfied		
ω_7	1	1	1	satisfied	violated	satisfied

Table 1: Truth values of bird (b), flies (f), and penguin (p) for each world ω_i, and whether each world satisfies or violates the default rules.

Based on the constraints, these worlds are ranked to reflect which interpretations are more plausible. In table 1 each line represents a world with the combination of the three predicates and the satisfaction or violation of a default rule. In this table it can be observed that no world satisfies all rules, which highlights the core problem. The worlds ω_0 and ω_2 do not satisfy or violate any rule.
The first default rule, $d_1 : b \to f$ ("Birds fly"), increases the plausibility of worlds where birds fly ($b \wedge f$) compared to those where birds do not fly ($b \wedge \neg f$). This rule affects the ranking between worlds ω_6, ω_7 (where the bird flies) and ω_4, ω_5 (where the bird does not fly).
The second default rule, $d_2 : p \to \neg f$ ("Penguins do not fly"), prioritizes worlds where penguins do not fly ($p \wedge \neg f$) over those where penguins fly ($p \wedge f$). This affects the relative plausibility of ω_5, ω_1 (where the penguin does not fly) compared to ω_7 and ω_3 (where the penguin flies).
The third default rule, $d_3 : p \to b$ ("Penguins are birds"), ranks the worlds ω_5, ω_7 higher than ω_1, ω_3.
We can insert the worlds in which the rule holds into the constraints, as these worlds represent the rule:

$$
\begin{aligned}
C_1 &: \max(\pi(\omega_6), \pi(\omega_7)) > \max(\pi(\omega_4), \pi(\omega_5)),\\
C_2 &: \max(\pi(\omega_5), \pi(\omega_1)) > \max(\pi(\omega_3), \pi(\omega_7)),\\
C_3 &: \max(\pi(\omega_5), \pi(\omega_7)) > \max(\pi(\omega_1), \pi(\omega_3)).
\end{aligned}
$$

5.0.4 Ranking of the possible Worlds

With the combination of the constraints it is possible to find out an ordering, where no constarint is broken. The graph 1 gives a hint about the possibility relations between the worlds.
For example the constraint C_1 says that either ω_6 (blue arrows) or ω_7 (red arrows) must be more possible than ω_4 and ω_5. The correct ordering is found if the worlds can be arranged in levels with the relations (arrows) while the arrows are only going down from

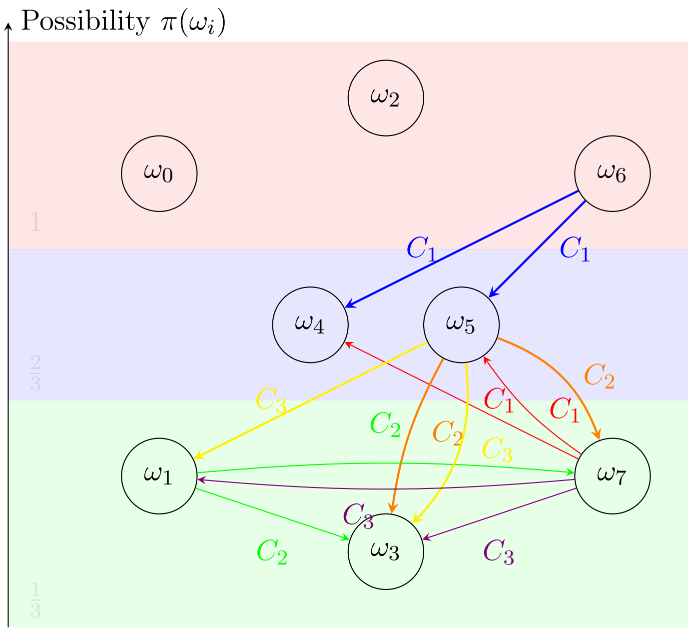

Figure 1: Possibility of the worlds ω_i and possibility relation between the worlds represented by the colored arrows

a higher possibility level to a lower one. In this example an ordering can be found with color groups blue (C_1), orange (C_2) and yellow (C_3). The ordering of the worlds threw the constraints results in:

$$\pi(\omega_0) = \pi(\omega_2) = \pi(\omega_6) = 1,$$
$$\pi(\omega_4) = \pi(\omega_5) = \frac{2}{3},$$
$$\pi(\omega_1) = \pi(\omega_3) = \pi(\omega_7) = \frac{1}{3}.$$

In this example the numerical representation is used for the different possibility levels of the worlds. The worlds ω_0 and ω_1 are in the highest possibility level, as no constraints grades the worlds down.

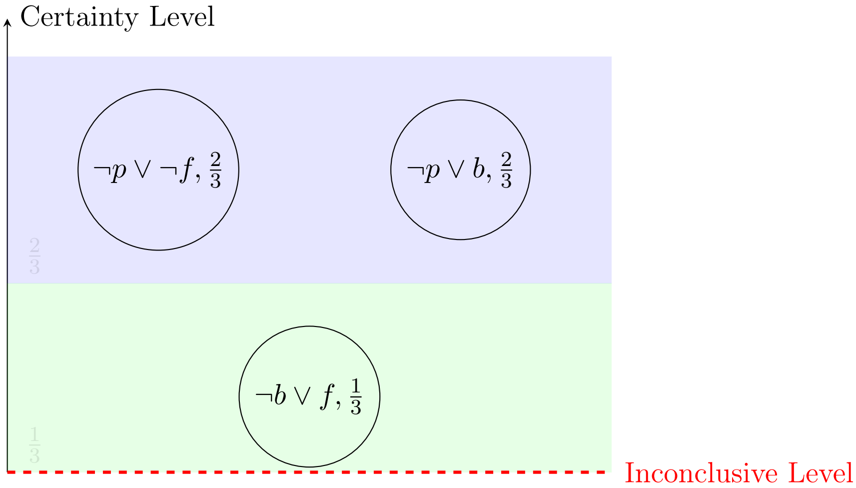

Figure 2: Possibilistic Knowledge Base of the Penguin Problem with the elements arranged over the certainty

5.0.5 Creating Knowledge Base

We calculate the necessity degrees from the possibility of the worlds as follows with the duality relationship of possibility and necessity:

$$\begin{aligned}
N(\neg p \vee \neg f) &= \min\{1-\pi(\omega) \mid \omega \models p \wedge f\} \\
&= \min(1-\pi(\omega_3), 1-\pi(\omega_7)) = \frac{2}{3} \\
N(\neg b \vee f) &= \min\{1-\pi(\omega) \mid \omega \models b \wedge \neg f\} \\
&= \min(1-\pi(\omega_4), 1-\pi(\omega_5)) = \frac{1}{3} \\
N(\neg p \vee b) &= \min\{1-\pi(\omega) \mid \omega \models p \wedge \neg b\} \\
&= \min(1-\pi(\omega_1), 1-\pi(\omega_3)) = \frac{2}{3}
\end{aligned}$$

Thus, we obtain the possibilistic knowledge base:

$$K = \{(\neg p \vee \neg f, \frac{2}{3}), (\neg p \vee b, \frac{2}{3}), (\neg b \vee f, \frac{1}{3})\}.$$

The possibilitic knowledge base is visualized in figure 2.

5.0.6 Reasoning with Uncertain Information

In this part we can find out if Tweety flies by introducing information about Tweety. First we introduce that Tweety is a bird:

$$F = \{(b, 1)\} \quad \text{(Tweety is a bird with certainty 1).}$$

From the rule $(\neg b \vee f, 1/3)$ in the knowledge base K, we can infer by using the weakest link resultion rule 4:

$$(\neg b \vee f, 1/3) \quad \text{and} \quad (b, 1) \vdash (f, \min(1, 1/3)) = (f, 1/3).$$

Thus, we conclude that Tweety flies with a certainty level of $\frac{1}{3}$. We can see the result graphically in figure 3. The violet arrow symbolizes the inference with the weakest link resolution rule.

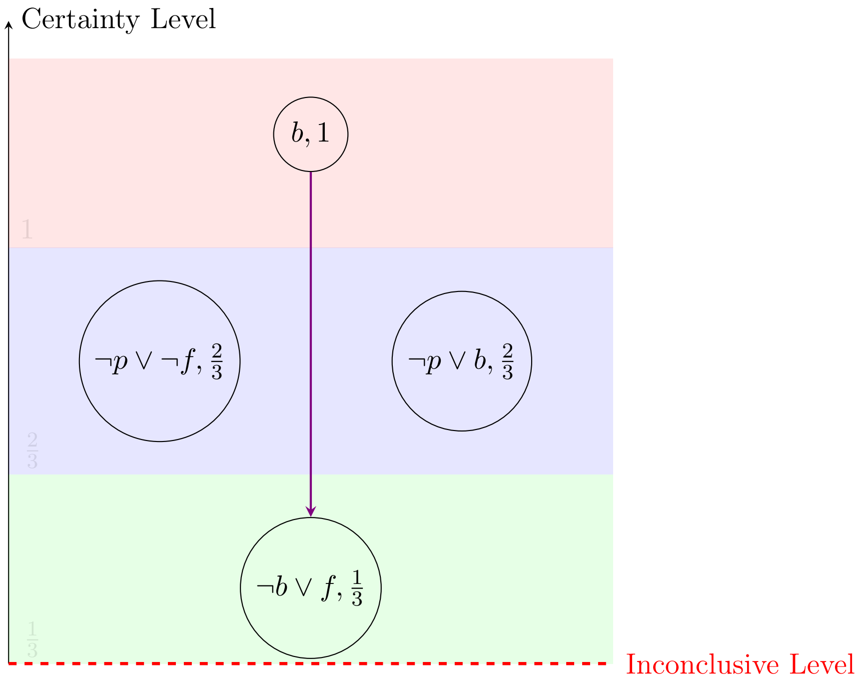

Figure 3: Possibilistic Knowledge Base of the Penguin Problem with first information about Tweety

If new information states that Tweety is also a penguin:

$$F = \{(b, 1), (p, 1)\} \quad \text{(Tweety is a penguin with certainty 1),}$$

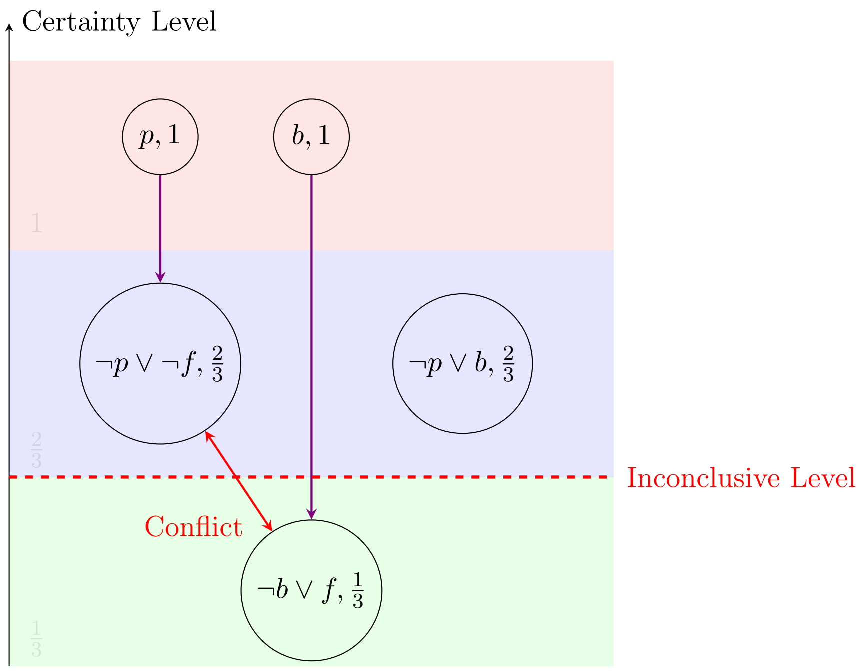

Figure 4: Possibilistic Knowledge Base of the Penguin Problem with new information about Tweety

then, using $(\neg p \vee \neg f, 2/3)$, we derive:

$$(\neg p \vee \neg f, 2/3) \quad \text{and} \quad (p, 1) \vdash (\neg f, \min(1, 2/3)) = (\neg f, 2/3).$$

Since we now have conflicting conclusions:

$$(\neg f, 2/3) \quad \text{and} \quad (f, 1/3) \vdash (\bot, \min(2/3, 1/3)) = (\bot, 1/3),$$

the system detects an inconsistency of level $\frac{1}{3}$. Given the principle that higher certainty information takes advantage, the inference that Tweety flies is drowned, leading to the final conclusion:

$$K \cup F \vdash (\neg f, 2/3),$$

which means *"Tweety does not fly"* with a certainty of $\frac{2}{3}$.
The results are visualized in the figure 4. It can be seen that the inconclusive level is raised to a level where above no conflict is between the information.

This example shows how possibilistic logic can handle inconsistency when new information is added. The calculation of the necessities of the rules out of the possible worlds is one key aspect in this example.

6 Discussion and Conclusion

References

[1] Didier Dubois, Jérôme Lang, and Henri Prade. A brief overview of possibilistic logic. In Rudolf Kruse and Pierre Siegel, editors, *Symbolic and Quantitative Approaches to Uncertainty: European Conference ECSQAU, Marseille, France, October 15-17, 1991. Proceedings*, volume 548 of *Lecture Notes in Computer Science book series (LNCS)*, pages 53–57, Marseilles, France, October 1991. Springer Berlin Heidelberg.

[2] Didier Dubois and Henri Prade. Possibilistic logic — an overview. In Jörg H. Siekmann, editor, *Computational Logic*, volume 9 of *Handbook of the History of Logic*, pages 283–342. North-Holland, 2014.

[3] Didier Dubois and Henri Prade. From possibilistic rule-based systems to machine learning-a discussion paper. In *Scalable Uncertainty Management: 14th International Conference, SUM 2020, Bozen-Bolzano, Italy, September 23–25, 2020, Proceedings 14*, pages 35–51. Springer, 2020.

[4] Didier Dubois and Henri Prade. Possibility theory and possibilistic logic: Tools for reasoning under and about incomplete information. In *Intelligence Science III: 4th IFIP TC 12 International Conference, ICIS 2020, Durgapur, India, February 24–27, 2021, Revised Selected Papers 4*, pages 79–89. Springer, 2021.

[5] Didier Dubois and Henri Prade. Possibility theory. In *Granular, Fuzzy, and Soft Computing*, pages 859–876. Springer, 2023.

[6] Didier Dubois, Henri Prade, and Steven Schockaert. Generalized possibilistic logic: foundations and applications to qualitative reasoning about uncertainty. *Artificial Intelligence*, 252:139–174, 2017.

[7] Michel Grabisch. Temporal scenario modelling and recognition based on possibilistic logic. *Artificial Intelligence*, 148(1):261–289, 2003. Fuzzy Set and Possibility Theory-Based Methods in Artificial Intelligence.

[8] Joseph Y. Halpern. An analysis of first-order logics of probability. *Artificial Intelligence*, 46(3):311–350, 1990.

[9] Bernhard Hollunder. An alternative proof method for possibilistic logic and its application to terminological logics. *International Journal of Approximate Reasoning*, 12(2):85–109, 1995.

[10] Jérôme Lang. Possibilistic logic: complexity and algorithms. In *Handbook of Defeasible Reasoning and Uncertainty Management Systems: Algorithms for Uncertainty and Defeasible Reasoning*, pages 179–220. Springer, 2000.

[11] Cosimo Damiano Persia. Learning possibilistic logic theories. 2023.

[12] Salem BENFERHAT-Didier DUBOIS-Henri PRADE. Representing default rules in possibilistic logic. In *3rd International conference on principles of knowledge representation and reasoning (KR'92)*, pages 673–684, 1992.

[13] L.A. Zadeh. Fuzzy sets. *Information and Control*, 8(3):338–353, 1965.

[14] L.A Zadeh. Fuzzy sets as a basis for a theory of possibility. *Fuzzy Sets and Systems*, 1(1):3–28, 1978.

Statement

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